Question ID: 22e4d633

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question

Although many transposons, DNA sequences that move within an organism's genome through shuffling or duplication, have become corrupted and inactive over time, those from the long interspersed nuclear elements (LINE) family appear to remain active in the genomes of some species. In humans, they are functionally important within the hippocampus, a brain structure that supports complex cognitive processes. When the results of molecular analysis of two species of octopus—an animal known for its intelligence—were announced in 2022, the confirmation of a LINE transposon in *Octopus vulgaris* and *Octopus bimaculoides* genomes prompted researchers to hypothesize that that transposon family is tied to a species' capacity for advanced cognition.

Which finding, if true, would most directly support the researchers' hypothesis?

Answer

A. The LINE transposon in *O. vulgaris* and *O. bimaculoides* genomes is active in an octopus brain structure that functions similarly to the human hippocampus.

B. The human genome contains multiple transposons from the LINE family that are all primarily active in the hippocampus.

C. A consistent number of copies of LINE transposons is present across the genomes of most octopus species, with few known corruptions.

D. *O. vulgaris* and *O. bimaculoides* have smaller brains than humans do relative to body size, but their genomes contain sequences from a wider variety of transposon families.

Correct Answer: A

Rationale

Choice A is the best answer. The text says that LINE transposons are important in the human hippocampus, which supports complex cognition. If the LINE transposon found in octopuses is active in a similar part of their brain, that would suggest that LINE transposons support complex cognition in octopuses too, which in turn supports the hypothesis that LINE transposons are linked to advanced cognition in general.

Choice B is incorrect. This choice doesn't support the hypothesis. It doesn't include anything about how LINE transposons function in species other than humans. Choice C is incorrect. This choice doesn't support the hypothesis. It doesn't include anything about how the LINE transposon in octopuses might support advanced cognition. Choice D is incorrect. This choice doesn't support the hypothesis. It doesn't include anything about how the LINE transposon in octopuses might support advanced cognition.